

Predicting the Mechanical Properties of Complex Lattice Metamaterials Using Graph Neural Networks

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Abstract: Lattice metamaterials are engineered solids whose mechanical performance is governed not only by constituent material properties but by geometry, topology, nodal connectivity, and defect structure. Classical analytical models assume ideal, periodic unit cells, yet real fabricated lattices produced via additive manufacturing exhibit strut diameter variation, node misalignment, and missing members that cause these models to fail. Finite element (FE) simulation can resolve these complexities but becomes computationally prohibitive when screening large design spaces across thousands of topologies and defect configurations. This work addresses the need for a fast, topology-aware surrogate model by framing lattice stiffness and fracture-related property prediction as a graph learning problem. A lattice is naturally represented as a graph in which joints are nodes and struts are edges, with node and edge attributes encoding position, beam cross-section, and boundary conditions. Graph Neural Networks (GNNs) operate natively on this representation and can propagate information along connected struts in a manner that mirrors physical load-path transmission through the structure. The research methodology proceeds in two phases. First, a synthetic dataset is generated using a custom Abaqus scripting-interface pipeline that produces randomized 2-D Voronoi-like beam-lattice geometries. The pipeline sweeps a randomness parameter from 0.0 (regular hexagonal) to 0.5 (highly disordered), running 3,000 static FE simulations and exporting node features, adjacency information, global stiffness, and element stress fields as structured CSV data for each sample. Second, a two-layer Graph Convolutional Network (GCN) is trained on the resulting dataset using an 80/10/10 train-validation-test split, with two-phase learning rate scheduling (exploration at high learning rate followed by fine-tuning) and batch normalization for training stability. The model takes node coordinate features and graph connectivity as input and regresses global lattice stiffness as the primary target. Preliminary results demonstrate that the GNN learns meaningful structure-property relationships from topology alone, with current validation $R^2 \approx 0.675$. Near-term work focuses on incorporating richer edge attributes (beam length, cross-section area, orientation angle) and extended graph convolution depth to improve predictive accuracy. Longer-term goals include extending the model to predict fracture-relevant quantities such as mode-I stress intensity factor and T-stress contribution, validating against Fleck-Deshpande-Ashby scaling laws, and testing generalization to unseen topology families (kagome, triangular, octet-truss). This approach has the potential to reduce the cost of lattice design iteration by orders of magnitude relative to full FE solves, enabling defect-aware optimization of metamaterials for aerospace, biomedical, and energy absorption applications.